

Vector Database Infrastructure and Indexing Benchmarks

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1. Vector Store Implementations

MCPPRO defines an abstract base interface BaseVectorStore with 4 production backends:

- InMemoryVectorStoreService: Pure Python in-memory index with disk serialization support for instantaneous cold-start restarts.
- QdrantVectorStoreService: Supports local embedded storage (:memory: or local directory) and remote gRPC cloud clusters. Uses Cosine distance metric with HNSW indexing.
- SupabaseVectorStoreService: Integrates PostgreSQL pgvector extension through custom RPC match_documents functions.
- PineconeVectorStoreService: Serverless managed index on AWS us-east-1.

2. Performance Targets

- Batch Insertion Latency: Target < 120ms per 100 vectors
- Top-10 Retrieval Latency: Target < 15ms in-memory, < 45ms over Qdrant gRPC
- Memory Footprint: InMemory consumes approximately 1.8MB RAM per 1,000 vectors with 1536 dimensions.

3. Deletion and Namespace Isolation

Each document processing run executes a complete vector space purge or namespace filter to guarantee zero data bleeding across concurrent user sessions.